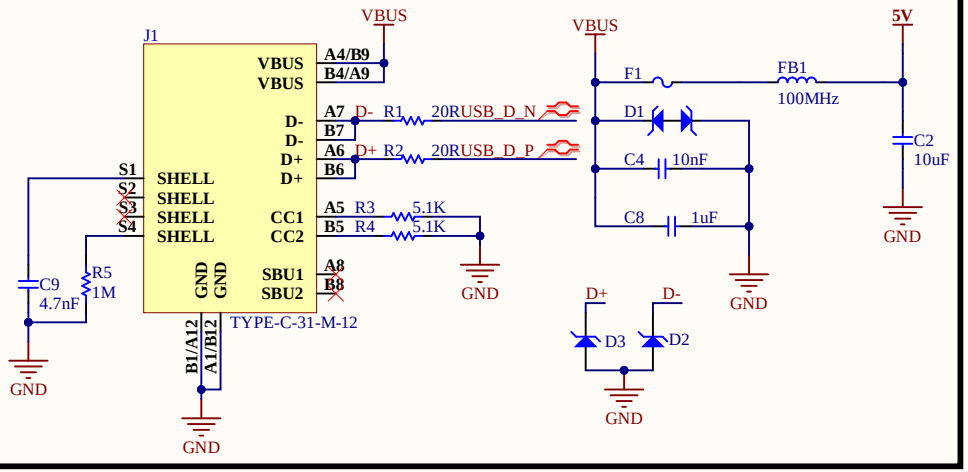
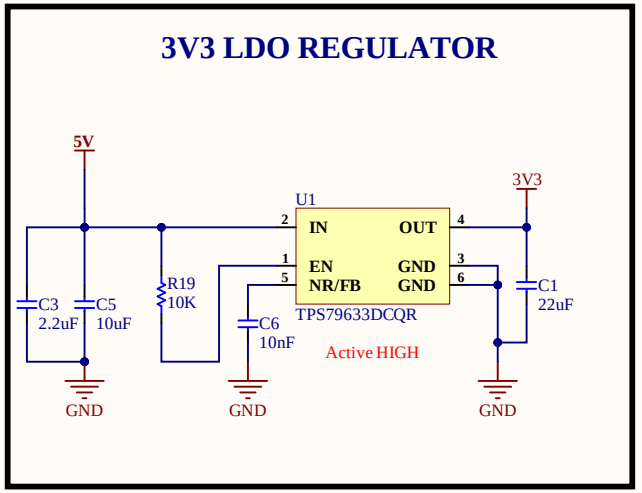
VBUS 5V INPUT



3V3 LDO REGULATOR



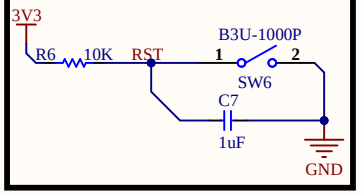
USB_D_N
USB_D_P

5V
3V3

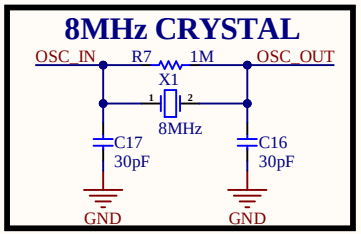
GND

Title: Power_Sheet.SchDoc	
File: Bento Pad Mini-v1.0.PrjPcb	
Date: 13/05/2025	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 3

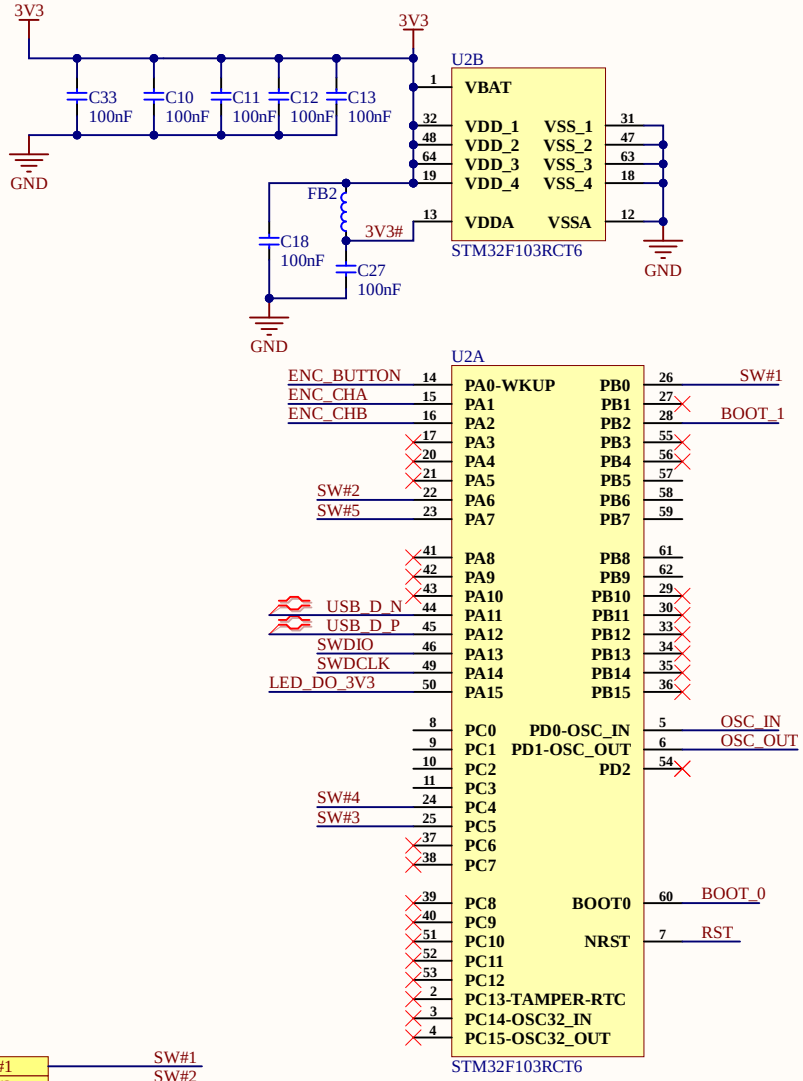
RESET



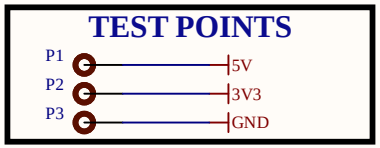
8MHz CRYSTAL



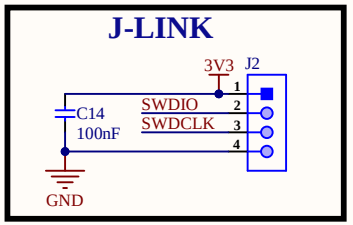
MCU



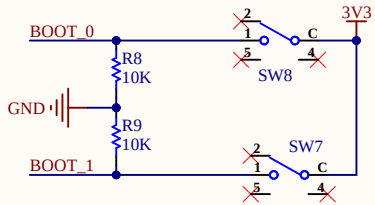
TEST POINTS

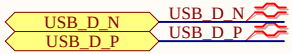


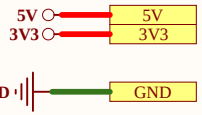
J-LINK

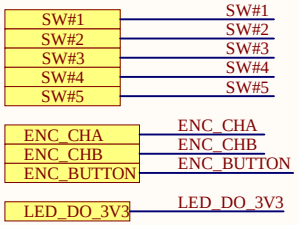


BOOT_MODE SELECT

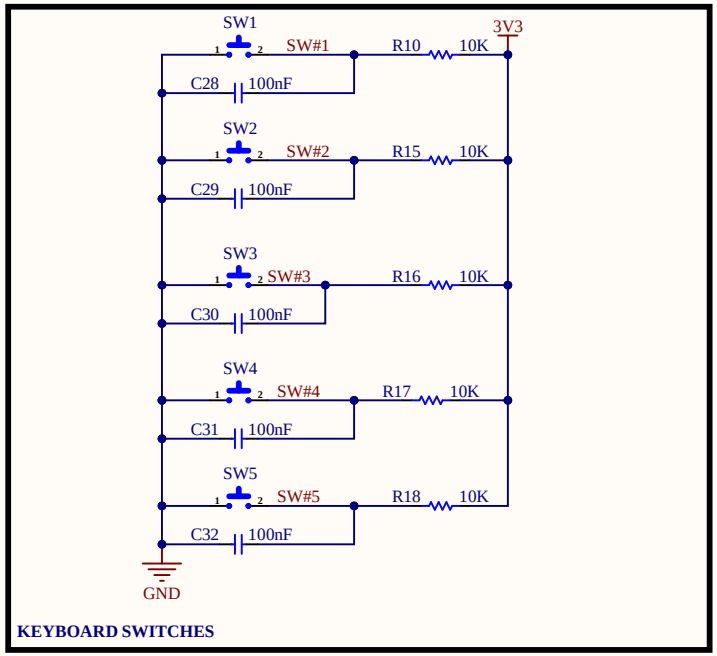


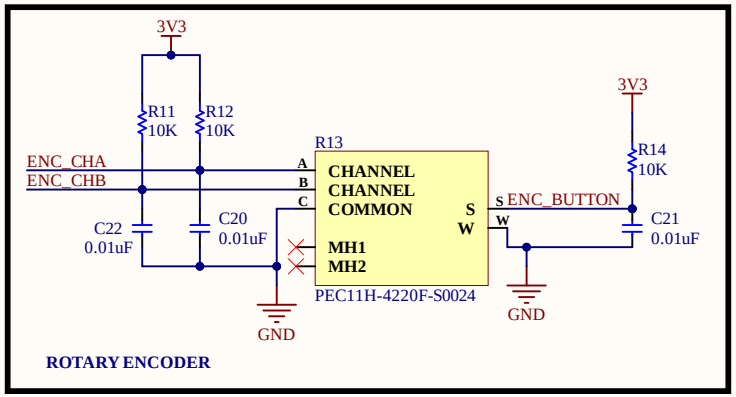


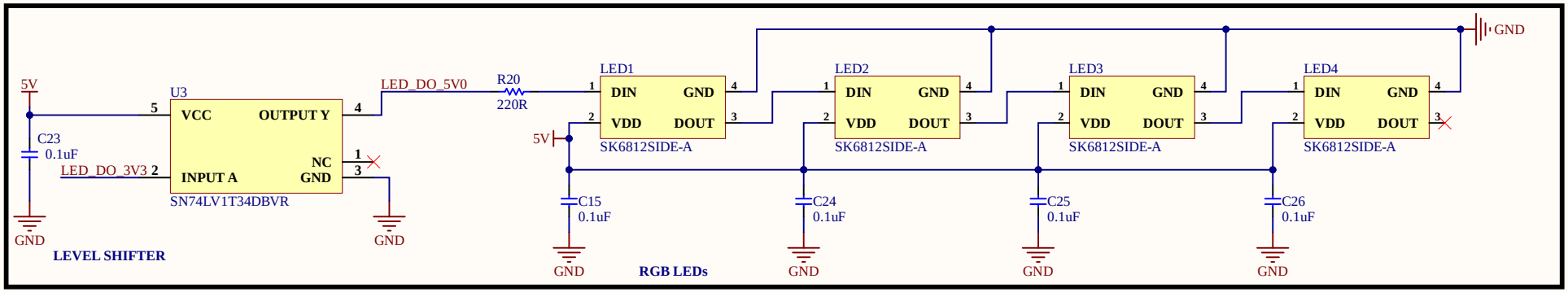


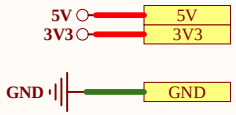


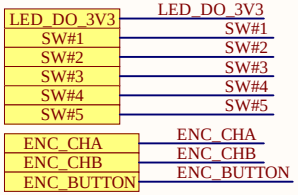
Title: Controller_Sheet.SchDoc	
File: Bento Pad Mini-v1.0.PrjPcb	
Date: 12/05/2025	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 3











eSpark Consulting Group

Title: Button Sheet.SchDoc	
File: Bento Pad Mini-v1.0.PrjPcb	
Date: 12/05/2025	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 3

Comment	Description	Designator	Footprint	LibRef	Quantity
22uF	CAP CER 22UF 10V X5R 0603	C1	Cap-0603	CC0603MRX5R6BB226	1
10uF	CAP CER 10UF 25V X5R 0603	C2	Cap-0603	C1608X5R1E106M080AC	1
2.2uF	CAP CER 2.2UF 25V X5R 0402	C3	Cap-0402	GRM155R61E225KE11D	1
10nF	CAP CER 10000PF 50V C0G/NPO 0603	C4	Cap-0603	CC0603JRNPO9BN103	1
10uF	CAP CER 10UF 10V X5R 0402	C5	Cap-0402	CC0402MRX5R6BB106	1
10nF	CAP CER 10000PF 50V X7R 0402	C6	Cap-0402	CC0402KRX7R9BB103	1
1uF	CAP CER 1UF 35V X5R 0402, CAP CER 1UF 16V X5R 0402	C7, C8	Cap-0402	C1005X5R1V105K050BC, CC0402KRX5R7BB105	2
4.7nF	CAP CER 4700PF 50V X7R 0402	C9	Cap-0402	CC0402KRX7R9BB472	1
100nF	CAP CER 0.1UF 50V X7R 0402	C10, C11, C12, C13, C14, C18, C27, C28, C29, C30, C31, C32, C33	cap-0402	C1005X7R1H104KT000F	13
0.1uF	CAP CER 0.1UF 25V X7R 0402	C15, C23, C24, C25, C26	Cap-0402	GRM155R71E104KE14D	5
30pF	CAP CER 30PF 50V C0G/NPO 0402	C16, C17	Cap-0402	CC0402GRNPO9BN300	2
0.01uF	CAP CER 10000PF 25V X5R 0402	C20, C21, C22	Cap-0402	CC0402KRX5R8BB103	3
PESD5V0H1BSFYL	TVS DIODE 5VWM 6.3VC DSN0603-2	D1	DSN0603-2	PESD5V0H1BSFYL	1
ESD9L5.0ST5G	TVS DIODE 5V 9.8V SOD923	D2, D3	SOD-923.	ESD9L5.0ST5G	2
BSMD0805L-200	6V 2A 50A 1W -40℃~+85℃ 4A 4mΩ 5s 45mΩ 0805 Resettable Fuses ROHS	F1	Fuse-0805	BSMD0805L-200	1
100MHz	FERRITE BEAD 120 OHM 1206 1LN	FB1	IND-1206	BLM31PG121SN1L	1
MMZ2012R102ATD25	FERRITE BEAD 1K OHM 0805 1LN	FB2	Ind-0805	MMZ2012R102ATD25	1
TYPE-C-31-M-12	5A 1 16P Female Type-C SMD USB Connectors ROHS	J1	TYPE-C-31-M-12_	TYPE-C-31-M-12	1
X6511WV-04H-C60D30	Gold 250V 3A Straight Square Pins 2.5mm 4P 260℃ 6mm -40℃~+105℃ 3mm 2.54mm Plugin Black Brass 1x4P Plugin,P=2.54mm Pin Headers ROHS	J2	HDR1X4_	X6511WV-04H-C60D30	1
SK6812SIDE-A	SMD-4P,1.6x4mm Light Emitting Diodes (LED) ROHS	LED1, LED2, LED3, LED4	SK6812SIDE-A	SK6812SIDE-A	4
TEST POINT	Test Point ON	P1, P2, P3	Test-point	TEST POINT	3
20R	RES SMD 20 OHM 0.1% 1/16W 0402	R1, R2	Res-0402	RT0402BRD0720RL	2
5.1K	RES 5.1K OHM 1% 1/16W 0402	R3, R4	Res-0402	RC0402FR-075K1L	2
1M	RES SMD 1M OHM 5% 1/16W 0402, RES 1M OHM 1% 1/16W 0402	R5, R7	Res-0402	AF0402JR-071ML, RC0402FR-071ML	2
10K	RES 10K OHM 1% 1/16W 0402	R6, R8, R9, R10, R11, R12, R14, R15, R16, R17, R18, R19	Res-0402	RC0402FR-0710KL	12
PEC11H-4220F-S0024	11mm Encoder with 24 Detent Force 20mm Metal Flatted Shaft PC Pin Horizontal/Rear Facing	R13	PEC11H-4220F-S0024	PEC11H-4220F-S0024	1
220R	RES SMD 220 OHM 1% 1/10W 0603	R20	Res-0603	RC0603FR-07220RL	1
CPG151101S11-16	SMD,14.5x5.9mm Mechanical Keyboard Shaft ROHS	SW1, SW2, SW3, SW4, SW5	CPG151101S11-16	CPG151101S11-16	5
B3U-1000P	SWITCH TACTILE SPST-NO 0.05A 12V	SW6	B3U-1000P	B3U-1000P	1
MK-11C04-G013	Surface Mount 1A SPDT 6V Black 10000次 SMD Slide Switches ROHS	SW7, SW8	MK-11C04-G013	MK-11C04-G013	2
TPS79633DCQR	IC REG LINEAR 3.3V 1A SOT223-6	U1	SOT-223-6_	TPS79633DCQR	1
STM32F103RCT6	IC MCU 32BIT 256KB FLASH 64LQFP	U2	64-LQFP (10x10)	STM32F103RCT6	1
SN74LV1T34DBVR	IC BUF NON-INVERT 5.5V SOT23-5	U3	SOT-23-5	SN74LV1T34DBVR	1
8MHz	8MHz Surface Mount Crystal 20pF ±10ppm ±30ppm SMD5032 Crystals ROHS	X1	0153G2-8.000F20DTNJL	0153G2-8.000F20DTNJL	1